

# Ranked Retrieval 1: 从 match/no match 到排序分数

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

Boolean retrieval limitation → top-k ranked output → Jaccard baseline → TF/DF/IDF → TF-IDF weighting → Vector Space Model → length normalization → cosine similarity → SMART notation

10/7/25

## Boolean Retrieval

- Thus far, our queries have all been Boolean.
  - Documents either: “match” or “no match”.
- Good for expert users with precise understanding of their needs and the collection.
  - Patent search uses sophisticated sets of Boolean queries and check hundreds of search results  
(car OR vehicle) AND (motor OR engine) AND NOT (cooler)
- Not good for the majority of users.
  - Most incapable of writing Boolean queries.
  - Most don't want to go through 1000s of results.
    - This is particularly true for web search
    - Question: What is the most unused web-search feature?

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## Ranked Retrieval

- Typical queries: free text queries
- Results are “ranked” with respect to a query
- Large result sets are not an issue
  - We just show the top  $k$  ( $\approx 10$ ) results
  - We don't overwhelm the user
- Criteria:
  - Top ranked documents are the most likely to satisfy user's query
  - Score is based on how well documents match a query  
**Score( $d, q$ )**

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## 这节课一句话

这节课把检索从 Boolean 的“进不进名单”，升级为 ranked retrieval 的“谁最应该排在前面”。

# 1. Boolean retrieval 的天花板

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### Boolean Retrieval

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### Ranked Retrieval

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## 人话直觉

Boolean 像门口保安：只说“能进/不能进”；但搜索引擎需要前台排序：谁最值得先见。

精确定义 / 机制： Boolean retrieval 给每个 document match/no match，适合专家用户和精确约束，但普通用户通常输入 free text query，并期待系统返回 top-k ranked results。ranked retrieval 给每个 document 一个 score，再按分数排序。

考试问法：若问 ranked vs Boolean，核心区别是 binary decision vs score-based ranking/top-k。

## 2. Ranking output: $\text{score}(d,q)$

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**Old Example**

- Find documents matching query {ink wink}
  - Load inverted lists for each query word
  - Merge two postings lists  $\rightarrow$  **Linear merge**
- Apply function for matches
  - Boolean: exist / not exist = 0 or 1
  - Ranked:  $f(tf, df, length, \dots) = 0 \rightarrow 1$

**Matches**

|             |               |     |     |     |
|-------------|---------------|-----|-----|-----|
| <b>ink</b>  | $\Rightarrow$ | 3:1 | 4:1 | 5:1 |
| <b>wink</b> | $\Rightarrow$ | 1:1 | 5:1 |     |

1:  $f(0,1)$   
3:  $f(1,0)$   
4:  $f(1,0)$   
5:  $f(1,1)$

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**Function example: Jaccard coefficient**

- a commonly used measure of overlap of two sets  $A$  and  $B$
- $jaccard(A,B) = \frac{|A \cap B|}{|A \cup B|}$
- $jaccard(A,A) = 1$
- $jaccard(A,B) = 0$ , if  $A \cap B = 0$
- Example:
  - $D1 \cup D2 = \{\text{he, likes, to, wink, and, drink}\}$
  - $D1 \cap D2 = \{\text{he, likes, to, drink}\}$
  - $jaccard(D1,D2) = \frac{4}{6} = 0.6667$

D1: He likes to wink, he likes to drink  
D2: He likes to drink, and drink, and drink

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### 人话直觉

不是把所有候选文档倒在桌上，而是给每份卷子打分，从高到低发给用户。

精确定义 / 机制：ranked retrieval 通常输出  $query_i d$ 、 $document_i d$  和  $score$ 。 $score(d,q)$  表示 document  $d$  与 query  $q$  的匹配程度或相关性估计。系统可以只返回 top-k，因为用户一般只看前几条。

$$d_1 > d_2 \quad \text{if} \quad score(q, d_1) > score(q, d_2)$$

考试问法：答 ranked retrieval pipeline 时要出现 score、sort、top-k 三个词。

### 3. Jaccard: 一个太朴素的 overlap 分数

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**Jaccard coefficient: Issues**

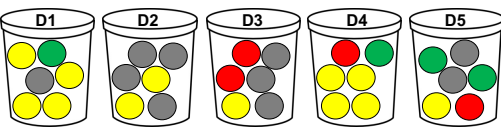
- Does not consider **term frequency** (how many times a term occurs in a document)
- It treats all terms equally!
  - How about **rare terms** in a collection? more informative than frequent terms.
  - *He likes to drink*, shall "to" == "drink"?
- Needs more sophisticated way of **length** normalization
  - $|D1| = 3, |D2| = 1000!$
  - $D1 \rightarrow Q, D2 \rightarrow D$

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**Should terms be treated the same?**

- Collection of 5 documents (balls = terms)
- Query   
- Which is the least relevant document?
- Which is the most relevant document?



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#### 人话直觉

Jaccard 只数两袋词有多少交集，不管“the”和“serendipity”谁更珍贵，也不管某个词出现多少次。

精确定义 / 机制：Jaccard coefficient 衡量两个集合 overlap，取 intersection 除以 union。它能作为简单 similarity，但不考虑 term frequency，所有 terms 权重相同，也缺少良好的 length normalization。

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

考试问法：若问 Jaccard limitations，答不考虑 tf、不区分 rare/common terms、length normalization 弱。

## 4. TF: 一个词在本文档里有多响

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**TFIDF**

- **TFIDF:**  
Term Frequency, Inverse Document Frequency
- **$tf(t,d)$ :**  
number of times term  $t$  appeared in document  $d$ 
  - As  $tf(t,d) \uparrow \rightarrow$  importance of  $t$  in  $d \uparrow \uparrow$
  - Document about IR, contains "retrieval" more than others
- **$df(t)$ :**  
number of documents term  $t$  appeared in
  - As  $df(t) \uparrow \rightarrow$  importance if  $t$  in a collection  $\downarrow \downarrow$ 
    - "the" appears in many document  $\rightarrow$  not important
    - "FT" is not important word in financial times articles

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**DF, CF, & IDF**

- **DF  $\neq$  CF** (collection frequency)
  - $cf(t)$  = total number of occurrences of term  $t$  in a collection
  - $df(t) \leq N$  ( $N$ : number of documents in a collection)
  - $cf(t)$  can be  $\geq N$
- **DF** is more commonly used in IR than **CF**
  - **CF** is still used
- **$idf(t)$ :** inverse of  $df(t)$ 
  - As  $idf(t) \uparrow \rightarrow$  rare term  $\rightarrow$  importance  $\uparrow \uparrow$
  - $idf(t) \rightarrow$  measure of the informativeness of  $t$

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### 人话直觉

如果一篇文章反复提到 indexing，它大概率真的在讲 indexing；但第 10 次出现没有第 1 次那么惊艳。

精确定义 / 机制：term frequency  $tf(t,d)$  是 term  $t$  在 document  $d$  中出现次数。TF-IDF 常用 log-scaled  $tf$ ，让第一次出现贡献很大，后续出现仍加分但边际收益递减。

$$tf\text{-weight}(t,d) = 1 + \log_{10} tf(t,d)$$

考试问法：若问为什么用 log TF，答 dampen repeated occurrences，避免重复词无限加分。

## 5. DF/IDF: 越常见越不稀罕

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| DF vs CF |       |     |       |      |       |      |                                               |
|----------|-------|-----|-------|------|-------|------|-----------------------------------------------|
| he       | drink | ink | likes | pink | think | wink |                                               |
| 2        | 1     | 0   | 2     | 0    | 0     | 1    | ← D1: He likes to wink, he likes to drink     |
| 1        | 3     | 0   | 1     | 0    | 0     | 0    | ← D2: He likes to drink, and drink, and drink |
| 1        | 1     | 1   | 1     | 0    | 1     | 0    | ← D3: The thing he likes to drink is ink      |
| 1        | 1     | 1   | 1     | 1    | 0     | 0    | ← D4: The ink he likes to drink is pink       |
| 1        | 1     | 1   | 1     | 1    | 0     | 1    | ← D5: He likes to wink, and drink pink ink    |
| 5        | 5     | 3   | 5     | 2    | 1     | 2    | DF                                            |
| 6        | 7     | 3   | 6     | 2    | 1     | 2    | CF                                            |

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| IDF: formula                                        |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|----------|-----------|---|---|--------|-----|---|-----|-------|---|-----|--------|---|-------|---------|---|-----|-----------|---|--|
| $idf(t) = \log_{10} \left( \frac{N}{df(t)} \right)$ |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| • Log scale used to dampen the effect of IDF        |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| • Suppose $N = 1$ million →                         |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
|                                                     | <table><thead><tr><th>term</th><th><math>df(t)</math></th><th><math>idf(t)</math></th></tr></thead><tbody><tr><td>calpurnia</td><td>1</td><td>6</td></tr><tr><td>animal</td><td>100</td><td>4</td></tr><tr><td>sky</td><td>1,000</td><td>3</td></tr><tr><td>fly</td><td>10,000</td><td>2</td></tr><tr><td>under</td><td>100,000</td><td>1</td></tr><tr><td>the</td><td>1,000,000</td><td>0</td></tr></tbody></table> | term     | $df(t)$ | $idf(t)$ | calpurnia | 1 | 6 | animal | 100 | 4 | sky | 1,000 | 3 | fly | 10,000 | 2 | under | 100,000 | 1 | the | 1,000,000 | 0 |  |
| term                                                | $df(t)$                                                                                                                                                                                                                                                                                                                                                                                                              | $idf(t)$ |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| calpurnia                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                    | 6        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| animal                                              | 100                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| sky                                                 | 1,000                                                                                                                                                                                                                                                                                                                                                                                                                | 3        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| fly                                                 | 10,000                                                                                                                                                                                                                                                                                                                                                                                                               | 2        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| under                                               | 100,000                                                                                                                                                                                                                                                                                                                                                                                                              | 1        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |
| the                                                 | 1,000,000                                                                                                                                                                                                                                                                                                                                                                                                            | 0        |         |          |           |   |   |        |     |   |     |       |   |     |        |   |       |         |   |     |           |   |  |

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### 人话直觉

“the”像空气，哪里都有，不能帮你定位；稀有词像门牌号，越少见越能指路。

精确定义 / 机制：  $df(t)$  是包含 term  $t$  的 document 数量， $cf(t)$  是 term 在 collection 中出现总次数。  
inverse document frequency 用  $N/df(t)$  的  $\log$  衡量词的区分能力：出现在越多文档中的词 IDF 越低。

$$idf(t) = \log_{10} \frac{N}{df(t)}$$

考试问法：考试常让区分  $df$  与  $cf$ ： $df$  数 documents， $cf$  数 occurrences。

6. TF-IDF: 本文档高频 × 全集合稀有

10/7/25

TFIDF term weighting

- One the best known term weights schemes in IR
  - Increases with the number of occurrences within a document
  - Increases with the rarity of the term in the collection
- Combines TF and IDF to find the weight of terms

$$w_{t,d} = (1 + \log_{10}tf(t,d)) \times \log_{10}(\frac{N}{df(t)})$$

- For a query  $q$  and document  $d$ , retrieval score  $f(q,d)$ :

$$Score(q,d) = \sum_{t \in q \cap d} w_{t,d}$$

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Document/Term vectors with tfidf

|           | Antony and Cleopatra | Julius Caesar | The Tempest | Hamlet | Othello | Macbeth |
|-----------|----------------------|---------------|-------------|--------|---------|---------|
| Antony    | 5.25                 | 3.18          | 0           | 0      | 0       | 0.35    |
| Brutus    | 1.21                 | 6.1           | 0           | 1      | 0       | 0       |
| Caesar    | 8.59                 | 2.54          | 0           | 1.51   | 0.25    | 0       |
| Calpurnia | 0                    | 1.54          | 0           | 0      | 0       | 0       |
| Cleopatra | 2.85                 | 0             | 0           | 0      | 0       | 0       |
| mercy     | 1.51                 | 0             | 1.9         | 0.12   | 5.25    | 0.88    |
| worser    | 1.37                 | 0             | 0.11        | 4.15   | 0.25    | 1.95    |

→ Vector Space Model

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人话直觉

好关键词要同时满足两件事：在这篇文档里声音大，在别的文档里不吵。

精确定义 / 机制： TF-IDF term weight 同时奖励 document 内高频和 collection 中稀有。它是经典强 baseline，简单、可解释、易实现。query-document score 可对共有 query terms 的权重求和，也可作为 VSM 中的向量权重。

$$w_{t,d} = (1 + \log_{10}tf(t,d)) \log_{10} \frac{N}{df(t)}$$

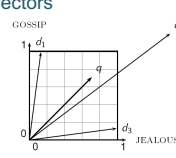
考试问法：若考 TF-IDF，必须定义 tf、df/idf，再解释“文档内多、集合中少”这一直觉。

## 7. Vector Space Model: 把文本放进几何空间

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**Vector Space Model**

- Documents and Queries are presented as vectors
- Match (Q,D) = Distance between vectors
- Example: Q= Gossip Jealous
- Euclidean Distance?  
*Distance between the endpoints of the two vectors*
- Large for vectors of diff. lengths
- Take a document d and append it to itself. Call this document d'.
  - "Semantically" d and d' have the same content
  - Euclidean distance can be quite large



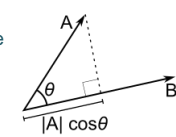
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**Angle Instead of Distance**

- The angle between the two documents is 0, corresponding to maximal similarity.
- Key idea: Rank documents according to angle with query.
  - Rank documents in increasing order of the angle with query
  - Rank documents in decreasing order of cosine (query, document)
- Cosine of angle = projection of one vector on the other



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### 人话直觉

query 和 document 都变成箭头；相关性不是看箭头绝对多长，而是看它们指向是不是相近。

**精确定义 / 机制：**VSM 将 documents 和 queries 表示为同一 term space 中的 weighted vectors。每个维度是 term，值可由 TF-IDF 等 weighting 给出。匹配时比较 query vector 与 document vector 的相似度。

**考试问法：**答 VSM 时要说 shared term space、weighted vectors、similarity/ranking。



## 8. Length normalization: 别奖励“啰嗦”

10/7/25

**Length Normalization**

- A vector can be normalized by dividing each of its components by its length – for this we use the  $L_2$  norm:

$$\|\vec{x}\|_2 = \sqrt{\sum_i x_i^2}$$

- Dividing a vector by its  $L_2$  norm makes it a unit (length) vector (on surface of unit hypersphere)
- Effect on the two documents  $d$  and  $d'$  ( $d$  appended to itself) from earlier slide: they have identical vectors after length-normalization.
  - Long and short documents now have comparable weights

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**Example**

- $D1 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \rightarrow \|\vec{D1}\|_2 = \sqrt{1 + 9 + 4} = 3.74$
- $D1_{normalized} = \begin{bmatrix} 0.267 \\ 0.802 \\ 0.535 \end{bmatrix}$
- $D2 = \begin{bmatrix} 3 \\ 9 \\ 6 \end{bmatrix} \rightarrow \|\vec{D2}\|_2 = \sqrt{9 + 81 + 36} = 11.25$
- $D2_{normalized} = \begin{bmatrix} 0.267 \\ 0.802 \\ 0.535 \end{bmatrix}$

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### 人话直觉

一篇文档复制粘贴一遍，语义没变，但长度翻倍；如果直接按点积或欧氏距离，就会被长度骗。

精确定义 / 机制：长文档包含更多 terms，容易得到更高 raw score。L2 normalization 将向量除以其长度，使比较更关注方向而非规模。复制文档后向量长度变大，但 normalized direction 基本不变。

$$\|\vec{x}\|_2 = \sqrt{\sum_i x_i^2}, \quad \vec{x}' = \frac{\vec{x}}{\|\vec{x}\|_2}$$

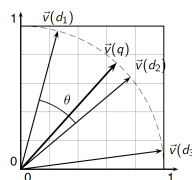
考试问法：若问为什么不用 Euclidean distance，答 document length 会扭曲相似度；cosine/normalization 更稳。

## 9. Cosine similarity: 看夹角，不看体重

10/7/25

**Cosine “Similarity” (Query, Document)**

- $\vec{q}_i$  is the tf-idf weight of term  $i$  in the query
- $\vec{d}_i$  is the tf-idf weight of term  $i$  in the document
- For normalized vectors:  
 $\cos(\vec{q}, \vec{d}) = \vec{q} \cdot \vec{d} = \sum_{i=1}^{|V|} q_i d_i$
- For non-normalized vectors:  
$$\cos(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \|\vec{d}\|} = \frac{\vec{q}}{\|\vec{q}\|} \cdot \frac{\vec{d}}{\|\vec{d}\|} = \frac{\sum_{i=1}^{|V|} q_i d_i}{\sqrt{\sum_{i=1}^{|V|} q_i^2} \sqrt{\sum_{i=1}^{|V|} d_i^2}}$$



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**Algorithm**

```
COSINESCORE( $q$ )
1  float Scores[ $N$ ] = 0
2  float Length[ $N$ ]
3  for each query term  $t$ 
4  do calculate  $w_{t,q}$  and fetch postings list for  $t$ 
5  for each pair( $d, \text{tf}_{t,d}$ ) in postings list
6  do Scores[ $d$ ] +=  $w_{t,d} \times w_{t,q}$ 
7  Read the array Length
8  for each  $d$ 
9  do Scores[ $d$ ] = Scores[ $d$ ] / Length[ $d$ ]
10 return Top  $K$  components of Scores[]
```

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### 人话直觉

两个箭头方向越一致，说明 query 想要的词权重模式和 document 越接近。

**精确定义 / 机制：**cosine similarity 是 query vector 与 document vector 的点积除以各自长度。若 vectors 已归一化，cosine 就等于 dot product。它是 VSM 中最常见的 ranking score。

$$\cos(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \|\vec{d}\|}$$

**考试问法：**公式题要写 numerator 是 dot product，denominator 是两个 L2 norms；归一化后直接点积。

## 10. SMART notation: 给 weighting scheme 起缩写名

10/7/25

**TFIDF Variants**

| Term frequency                                                                              | Document frequency                                     | Normalization                                   |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|
| n (natural) $tf_{t,d}$                                                                      | n (no) $1$                                             | n (none) $1$                                    |
| l (logarithm) $1 + \log(tf_{t,d})$                                                          | t (idf) $\log \frac{N}{df_t}$                          | c (cosine) $\frac{1}{\sqrt{w_q^2 + w_d^2}}$     |
| a (augmented) $0.5 + \frac{0.5 \cdot tf_{t,d}}{\max(tf_{t,d})}$                             | p (prob idf) $\max(0, \log \frac{N - df_t + 1}{df_t})$ | u (pivoted unique) $1/n$                        |
| b (boolean) $\begin{cases} 1 & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$ |                                                        | b (byte size) $1/CharLength^\alpha, \alpha < 1$ |
| L (log ave) $\frac{1 + \log(tf_{t,d})}{1 + \log(\max_{t \in d}(tf_{t,d}))}$                 |                                                        |                                                 |

- Many search engines allow for different weightings for queries vs. documents
- SMART** Notation: use notation *ddd.qqq*, using the acronyms from the table
- A very standard weighting scheme is: *Inc.Itc*

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**For Lab and CW**

| Term frequency                                                                              | Document frequency                                     | Normalization                                   |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|
| n (natural) $tf_{t,d}$                                                                      | n (no) $1$                                             | n (none) $1$                                    |
| l (logarithm) $1 + \log(tf_{t,d})$                                                          | t (idf) $\log \frac{N}{df_t}$                          | c (cosine) $\frac{1}{\sqrt{w_q^2 + w_d^2}}$     |
| a (augmented) $0.5 + \frac{0.5 \cdot tf_{t,d}}{\max(tf_{t,d})}$                             | p (prob idf) $\max(0, \log \frac{N - df_t + 1}{df_t})$ | u (pivoted unique) $1/n$                        |
| b (boolean) $\begin{cases} 1 & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$ |                                                        | b (byte size) $1/CharLength^\alpha, \alpha < 1$ |
| L (log ave) $\frac{1 + \log(tf_{t,d})}{1 + \log(\max_{t \in d}(tf_{t,d}))}$                 |                                                        |                                                 |

**“OR” operator, then:**

$$Score(q, d) = \sum_{t \in q \cap d} (1 + \log_{10} tf(t, d)) \times \log_{10} \left( \frac{N}{df(t)} \right)$$

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### 人话直觉

SMART 像配方标签：document 怎么处理、query 怎么处理，用六个字母浓缩写清楚。

**精确定义 / 机制：** SMART notation 用 *ddd.qqq* 描述 document 和 query 的 term weighting scheme，包括 tf scaling、idf usage、normalization 等。例如 *Inc.Itc* 表示 documents 和 queries 使用不同组合。它帮助系统化比较 weighting variants。

**考试问法：**若问 SMART，不必背所有组合细节，但要知道它描述 query/document 的 weighting scheme。

### 考试速记版

- Ranked retrieval 给 document 打 score 并返回 top-k；Boolean 只有 match/no match。
- Jaccard 是 |交集|/|并集|，缺点是不看 tf、idf 和长度。
- TF 表示词在文档中出现次数；log TF 体现边际收益递减。

- IDF 表示词的集合稀有性：df 越大，idf 越小。
- TF-IDF = 文档内高频 × 集合中稀有，是强 baseline。
- VSM 把 query/document 表示为 term vectors，用 cosine similarity 排序。
- Length normalization 防止长文档仅因更长而占便宜。

#### 一段稳妥考场回答

Ranked retrieval 解决 Boolean retrieval 只有 match/no match、不能排序的问题。系统为每个 document 对 query 计算 score，并返回 top-k。最简单的 Jaccard 只看集合 overlap，但不考虑 term frequency、term rarity 和 length normalization。TF-IDF 用 tf 衡量 term 在文档中的重要性，用 idf 衡量它在 collection 中的区分能力，常写为  $(1+\log tf)\log(N/df)$ 。在 Vector Space Model 中，query 和 document 都表示为 TF-IDF weighted vectors，再用 cosine similarity 比较方向相似度；L2 normalization 可减少文档长度差异造成的偏置。